

FA25-CSE-60641 - Project Proposal

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Prism Store - An QUIC based Object Storage System

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Project Proposal: Prism Store - An QUIC based Object Storage System

Abstract

This project proposes the design and implementation of a novel object storage kernel called PrismStore, specifically engineered to evaluate and optimize network protocol performance in distributed storage systems. The research aims to comprehensively analyze the impact of modern network protocols HTTP/3 with QUIC, HTTP/2, and gRPC on both client-server communication and inter-node coordination in object storage.

a. Problem Statement

Current object storage systems predominantly rely on traditional HTTP/1.1 and HTTP/2 protocols for communication, potentially leaving significant performance improvements unexploited. The performance of distributed storage systems heavily depends on network efficiency and I/O operations. With the emergence of HTTP/3 (QUIC) and gRPC protocols, there exists an opportunity to enhance storage system performance through strategic protocol selection and configuration.

b. Primary Goal

Design and develop PrismStore, a custom object storage kernel that enables comprehensive evaluation of different network protocols' impact on storage system performance.

c. Secondary Objectives

- Establish performance benchmarks against existing solutions (MinIO)
- Identify optimal protocol combinations for different communication channels
- Provide empirical evidence for protocol selection in distributed storage systems
- Develop a reference implementation for future object storage research

d. Technical Approach

- **System Architecture**

PrismStore will implement a distributed object storage system with configurable network protocols for two distinct communication channels:

- Client-to-Server Communication: External client access to the storage system
- Inter-Node Communication: Internal coordination between worker and master nodes

- **Protocol Implementation**

All protocols will be implemented with TLS 1.3 encryption to ensure security consistency across comparisons:

- HTTP/3 with QUIC: Modern protocol with built-in multiplexing and congestion control
- HTTP/2: Multiplexed HTTP with server push capabilities
- gRPC: High-performance RPC framework utilizing HTTP/2 transport

- **Performance Evaluation Framework**

The evaluation will systematically test protocol combinations across both communication channels, creating a comprehensive performance matrix.

e. Methodology

- **Phase 1: Core System Development**

Develop the PrismStore kernel with modular network protocol support, ensuring consistent functionality across all protocol implementations.

- **Phase 2: Baseline Establishment**

Deploy PrismStore with HTTP/2 configuration and benchmark against MinIO object storage system to establish credibility and performance baselines.

- **Phase 3: Protocol Evaluation**

- Step 1: Client Communication Analysis Compare HTTP/2 vs. HTTP/3 for client-server interactions Measure throughput, latency, and resource utilization
- Step 2: Inter-Node Communication Analysis Evaluate protocol impact on worker-master node coordination Assess scalability and fault tolerance characteristics
- Step 3: Cross-Protocol Evaluation Test gRPC for both communication channels Analyze performance against HTTP-based protocols

- **Phase 4: Optimization and Configuration Selection**

Systematically evaluate all possible protocol combinations to identify the optimal configuration for different workload patterns and network conditions.

f. Expected Outcomes

- **Research Contributions:** Comprehensive performance analysis of modern network protocols in object storage systems Empirical evidence for protocol selection guidelines in distributed storage architectures Open-source PrismStore implementation as a research platform
- **Performance Insights:** Quantitative comparison of protocol efficiency in storage workloads Identification of optimal protocol combinations for different use cases Network overhead analysis and resource utilization patterns
- **Practical Impact:** Performance optimization strategies for existing object storage deployments Protocol selection framework for storage system architects Reference implementation for educational and research purposes

g. Evaluation Metrics

Primary Metrics

- **Throughput:** Data transfer rates for upload/download operations
- **Latency:** Response times for storage operations
- **Scalability:** Performance degradation with increasing node count
- **Resource Utilization:** CPU, memory, and network resource consumption

Secondary Metrics

- **Connection Overhead:** Protocol establishment and maintenance costs
- **Error Recovery:** Performance under network failure conditions
- **Concurrent Operations:** Multi-client access patterns

h. Timeline and Deliverables

The project will deliver a complete PrismStore implementation with comprehensive performance evaluation results, demonstrating the impact of network protocol selection on distributed object storage system performance. The final system configuration will represent the optimal protocol combination identified through systematic evaluation.

i. Significance

This research addresses a critical gap in understanding how modern network protocols can enhance distributed storage system performance. By providing empirical evidence and a reference implementation, the project will contribute valuable insights to both academic research and industry practice in distributed systems architecture.